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Probabilistic Artificial Intelligence
Questions Pack

August 12, 2022

Time limit: 120 minutes

Instructions. This pack contains all questions for the final exam. It contains the questions only. Please use the accompanying answer sheet to provide your answers by blackening out the corresponding squares. As the exam will be graded by a computer, please **make sure to do blacken out the whole square and do not use ticks or crosses**. *During* the exam you can use a **pencil** to fill out the squares as well as an **eraser** to edit your answers. *After* the exam is over, we will collect the questions pack and provide you with additional time to blacken out the squares on the answer sheet with a **black pen**. *Nothing* written on pages of the question pack will be collected or marked. **Only the separate answer sheet with the filled squares will be marked**.

Please make sure that your answer sheet is clean and all answers are clearly marked by filling the squares out completely. We reserve the right to classify answers as wrong without further consideration if the sheet is filled out ambiguously.

Collaboration on the exam is strictly forbidden. You are allowed a summary of *two* A4 pages and a simple, non-programmable calculator. The use of any other helping material or collaboration will lead to being excluded from the exam and subjected to disciplinary measures by the ETH Zurich disciplinary committee.

Question Types In this exam, you will encounter the following question types.

- **Multiple Choice questions with a single answer.**

  Multiple Choice questions have **exactly one** correct choice. Depending on the difficulty of the question **2, 3, or 4** points are awarded if answered correctly, and **zero points** are awarded if answered wrong or not attempted.

- **True Or False questions.**

  Each True Or False questions has a value of **1 point** if answered correctly, **0 points** if answered wrong or not attempted.

Not all questions need to be answered correctly to achieve the best grade. There are **no negative grades** so you are incentivized to attempt all questions.

For your examination, preferably print documents compiled from auto-multiple-choice.

1 Probability, Bayesian linear regression

1.1 Unknown noise variance

Consider the model:

$$y_k = x_k + \varepsilon_k,$$

where ε_k are i.i.d $\mathcal{N}\left(0, \sigma^2\right)$ distributed and $x_k, y_k \in \mathbb{R}$. We have a prior on the variance parameter $\sigma^2 \sim \mathrm{IG}(\alpha, \beta)$, $\alpha, \beta > 0$, where the pdf for a random variable distributed via $T \sim \mathrm{IG}(\alpha, \beta)$ is:

$$p_T(t) = \frac{\beta^\alpha}{\Gamma(\alpha)} \frac{1}{t^{\alpha+1}} e^{-\frac{\beta}{t}}.$$

Here $\Gamma(\cdot)$ is the gamma function. We observe data $\{(x_1, y_1), \ldots, (x_n, y_n)\}$.

Question 1 *(2 points)* What is the likelihood $p(y_{1:n}|\sigma^2, x_{1:n})$?

A $\prod_{i=1}^n \frac{1}{\sqrt{2\pi\varepsilon_i^2}} e^{-\frac{(y_i - x_i)^2}{2\sigma^2}}$

B $\prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - x_i)^2}{2\varepsilon_i^2}}$

C $\prod_{i=1}^n \frac{1}{\sqrt{2\pi\varepsilon_i^2}} e^{-\frac{(y_i - x_i)^2}{2\varepsilon_i^2}}$

D $\prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - x_i)^2}{2\sigma^2}}$

Solution:
We compute:

$$p(y_{1:n}|\sigma^2, x_{1:n}) = \prod_{i=1}^n p(y_i|\sigma^2, x_i) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - x_i)^2}{2\sigma^2}}.$$

Hence the correct answer is D.

Question 2 *(4 points)* What is the the posterior distribution of the variance parameter $p(\sigma^2|x_{1:n}, y_{1:n})$?

A $\mathrm{IG}\left(\alpha + \frac{1}{2}\sum_{i=1}^n (y_i - x_i)^2, \beta + \frac{n}{2}\right)$

B $\mathrm{IG}\left(\alpha, \beta + \frac{n}{2}\sum_{i=1}^n (y_i - x_i)^2\right)$

C $\mathrm{IG}\left(\alpha + n, \beta + \sum_{i=1}^n (y_i - x_i)^2\right)$

D $\mathrm{IG}\left(\alpha + \frac{n}{2}, \beta + \frac{1}{2}\sum_{i=1}^n (y_i - x_i)^2\right)$

Solution:
We compute:

$$\begin{aligned} p(\sigma^2|x_{1:n}, y_{1:n}) &\propto p(\sigma^2) p(y_{1:n}|x_{1:n}, \sigma^2) = p(\sigma^2) \prod_{i=1}^n p(y_i|x_i, \sigma^2) \\ &\propto \frac{1}{(\sigma^2)^{\alpha+1}} e^{-\frac{\beta}{\sigma^2}} \prod_{i=1}^n \frac{1}{(\sigma^2)^{\frac{1}{2}}} e^{-\frac{(y_i - x_i)^2}{2\sigma^2}} \\ &= \frac{1}{(\sigma^2)^{\alpha+\frac{n}{2}+1}} e^{-\frac{1}{\sigma^2}\left(\beta + \frac{1}{2}\sum_{i=1}^n (x_i - y_i)^2\right)}, \end{aligned}$$

from where we conclude that the right answer is D.

1.2 Short questions

Question 3 *(1 point)* To represent a Gaussian distribution in $\mathbb{R}^d$ we need in the worst case $\frac{d^2+3d}{2}$ parameters

A True B False

Solution:
True. To represent mean we need d parameters. To represent off diagonal values of covariance matrix we need $\binom{d}{2}$ parameters and to represent diagonal values we need d parameters. Therefore we need

$$d + \binom{d}{2} + d = \frac{d^2 + 3d}{2}$$

parameters.

Question 4 *(1 point)* Let X, Y be i.i.d $\mathcal{N}(0, 1)$, then $X + Y$ and $X - Y$ are independent.

A True B False

Solution:
True. Since (X, Y) is Gaussian, independence of variables $X+Y$ and $X-Y$ is equivalent to them being not correlated. We compute:

$$\begin{aligned}\operatorname{cov}(X+Y, X-Y) &= \operatorname{cov}(X,X) - \operatorname{cov}(X,Y) + \operatorname{cov}(Y,X) - \operatorname{cov}(Y,Y) \\ &= 1 - 0 + 0 - 1 = 0,\end{aligned}$$

where we used bi-linearity of covariance operator on the first step. Hence $X + Y$ and $X - Y$ are independent.

Question 5 *(1 point)* Among all distributions over the real numbers with known mean and variance, Gaussian distribution has the maximum entropy.

A True B False

Solution:
True. Taken from Tutorial 2, page 3 - "Among all distributions over the real numbers with known mean and variance, the Gaussian distribution has the maximum entropy."

Question 6 *(1 point)* Bayesian Linear Regression can be viewed as a Gaussian Process regression with zero mean function and linear kernel function.

A True B False

Solution:
True. Taken from Lecture slides Gaussian Processes, page 17.

Question 7 *(1 point)* In the setting of Online Bayesian Linear Regression where input $x \in \mathbb{R}^d$, output $y \in \mathbb{R}$ and weights $w \in \mathbb{R}^d$, we can compute update $p_t(w) = p(w|\{(x_i, y_i)\}_{i=1}^t)$ given p_{t-1} in $\mathcal{O}(d^2)$

A True B False

Solution:
True. Taken from Homework 1, Exercise 6b.

2 Gaussian Processes

2.1 GP regression with equidistant inputs

We are given a dataset $\mathcal{D} = \{(x_0, y_0), \dots (x_N, y_N)\}$ consisting of $N+1$ data points $x_i, y_i \in \mathbb{R}$, for $N \in \mathbb{N}$. We assume every data point comes from the model

$$y_i = f(x_i) + \varepsilon_i, \quad \varepsilon_i \overset{\text{i.i.d}}{\sim} \mathcal{N}\left(0, \sigma^2\right),$$

where f is an *unknown* function and σ is a *known* parameter. For each N, the x_i are located equidistantly on the interval $[0, 1]$, i.e. $x_i = i/N$. For this question, consider GP regression,

$$f \sim GP(0, k),$$

where $k(u, v) = \sigma_F^2 \exp\left(-\frac{(u-v)^2}{2l^2}\right)$.

Question 8 *(4 point)* For constant $N > 0$ and $l > 0$, at all training locations x_i, the variance of your **posterior** distribution $p(f(x_i)|\mathcal{D})$ will asymptotically go to zero if

A σ_F goes to l^2.
B σ_F goes to σ.
C σ_F goes to 0.
D None of the other choices apply.

Solution:
C. From the lecture, we know that the posterior variance is given as

$$k'(x, x) := k(x, x) - k_{x,\mathcal{D}}(\boldsymbol{K}_{\mathcal{D},\mathcal{D}} + \sigma^2 \boldsymbol{I})^{-1} k_{x,\mathcal{D}}^T.$$

By definition,

$$k(x_i, x_j) = \sigma_F^2 \exp\left(-\frac{|i-j|^2}{N^2 2l^2}\right).$$

Since $|i - j|$ is lower bounded by 0, we get $\lim_{\sigma_F \to 0} k(x_i, x_j) = 0$.
For both σ and l, $N = 1$ provides a simple counter-example.

Question 9 *(4 points)* For constant $N > 0$ and $\sigma_F > 0$, consider the limit where the lengthscale l approaches zero. At all training locations x_i, the variance of your posterior distribution $p(f(x_i)|\mathcal{D}, \sigma)$ will in the limit go to

A $\frac{\sigma_F^2+\sigma^2}{2}$.

B $\frac{\sigma_F^2\sigma^2}{\sigma_F^2+\sigma^2}$.

C σ_F^2.

D σ^2.

Solution:
B. From the lecture, we know that the posterior variance is given as

$$k'(x_t, x_t) \coloneqq k(x_t, x_t) - k_{x_t,\mathcal{D}}(\boldsymbol{K}_{\mathcal{D},\mathcal{D}} + \sigma^2\boldsymbol{I})^{-1}k_{x_t,\mathcal{D}}^T.$$

By definition,

$$k(x_i, x_j) = \sigma_F^2 \exp\left(-\frac{|i-j|^2}{2N^2l^2}\right).$$

Since $0 \leq \frac{|i-j|^2}{N^2} \leq 1$, we get $\lim_{l\to 0} k(x_i, x_j) = \delta(i-j)\sigma_F^2$, where δ is the dirac delta function, i.e.,

$$\delta(x) = \begin{cases} 1; & x = 0 \\ 0; & x \neq 0 \end{cases}$$

This allows us to calculate limits for all elements of the RHS of the posterior variance, i.e.

$$\lim_{l\to 0} k(x_t, x_t) = \sigma_F^2$$
$$\lim_{l\to 0} (k_{x_t,\mathcal{D}})_m = \sigma_F^2\delta(m-t)$$
$$\lim_{l\to 0} (\boldsymbol{K}_{\mathcal{D},\mathcal{D}} + \sigma^2\boldsymbol{I})^{-1} = (\sigma_F^2+\sigma^2)^{-1}\boldsymbol{I}$$

Combining the law of sums and the law of products for limits, we finally obtain

$$\begin{aligned}\lim_{l\to 0} k'(x_t, x_t) &= \sigma_F^2 - \frac{\sigma_F^4}{\sigma_F^2+\sigma_F^2} \\ &= \frac{\sigma_F^2\sigma^2}{\sigma_F^2+\sigma^2}.\end{aligned}$$

Question 10 *(2 point)* Consider the same set up as in the previous two questions. Consider replacing the kernel k with a different kernel. What can be said about the posterior distribution $p(f(x_i)|\mathcal{D})$?
The posterior distribution of a Gaussian process is...

A ... Gaussian if and only if the kernel you choose is linear.

B ... Gaussian if and only if the kernel you choose is shift-invariant.

C ... Gaussian if and only if you choose the RBF kernel.

D ... always Gaussian.

Solution:
When defining the posterior distribution of a GP in the lecture slides, the kernel was treated as a blackbox. Thus, the posterior distribution is always Gaussian, by definition of a Gaussian process.

2.2 Kernel Choice

We have 3 datasets illustrated in Figure 1. Each datasets consists of pairs $(x_i, y_i)_{i \in I}$ and for every dataset we would like to learn a mapping f such that $y = f(x)$. We decide to model functions f with Gaussian Processes, i.e., :

$$f \sim GP(0, k(x_1, x_2)).$$

We try 4 different kernels k which are shown in Table 1. For every kernel k ($k \in \{A, B, C, D\}$) and dataset i ($i \in \{1, 2, 3\}$) we find such a configuration of trainable parameters (trainable parameters from Table 1) that the **marginal log-likelihood** of the data is maximized. For every pair of kernel k and dataset i we store the largest marginal log-likelihood of the data into the table as depicted in Table 2. Unfortunately we forgot to name the kernels and we also forgot the order of the kernels in the training procedure. Please recover the kernels!

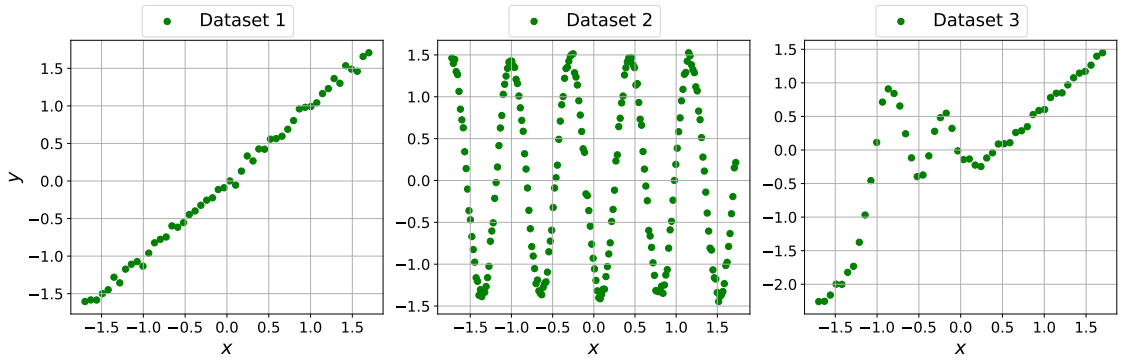


Figure 1: Different datasets

Table 1: Different kernels

Kernel	$k(u, v)$	Trainable parameters
A	uv	None
B	$e^{-\frac{\sin^2\left(\frac{\lvert u-v\rvert}{p}\right)}{2l^2}}$	l, p
C	$e^{-\frac{(u-v)^2}{2l^2}}$	l
D	$uv + e^{-\frac{(u-v)^2}{2l^2}}$	l

Table 2: **Marginal Log-likelihoods** of the data for different kernels

	Dataset 1	Dataset 2	Dataset 3
Kernel W	1.15	1.28	0.56
Kernel X	1.15	1.18	0.56
Kernel Y	1.36	-196.59	-70.31
Kernel Z	1.27	1.17	0.60

Question 11 *(1 point)* Kernel A is

A W B X C Y D Z

Solution:
Y. Kernel A is linear and since relation between x and y is linear in the dataset 1, the kernel has the largest log-likelihood among all kernels on this dataset.

Question 12 *(1 point)* Kernel B is

A W B X C Y D Z

Solution:
W. Kernel B is periodic and since relation between x and y is periodic in the dataset 2, the kernel has the largest log-likelihood among all kernels on this dataset.

Question 13 *(1 point)* Kernel C is

A W B X C Y D Z

Solution:
X. Since all other kernels were designed to be the best on one specific dataset, this kernel is the one that doesn't achieve the largest log-likelihood on any dataset.

Question 14 *(1 point)* Kernel D is

A W B X C Y D Z

Solution:
Z. Kernel D is linear plus RBF. Since dataset 3 has linear trend with some decaying sinusoids around this kernel should perform the best on dataset 3.

2.3 Hyperparameters of an RBF kernel

Again, we build a regression model using a Gaussian process $g \sim \mathcal{GP}(0, k)$. Your data set consists of one-dimensional features x, that should be mapped to one-dimensional targets. As kernel, k is chosen to be an RBF kernel, $k(x_i, x_j) = \sigma_F^2 \exp\left(-\frac{1}{2l^2}(x_i - x_j)^2\right)$ and the prior mean of your GP is 0. Before you apply this model, you standardize both your targets as well as your features, so that both are in an interval [0, 1]. For l, you are undecided which value you should choose.

Question 15 *(1 point)* You decide you want to use either $l = 0.8$ or $l = 0.1$. By choosing $l = 0.1$, what prior belief is expressed with this choice (as compared to 0.8)?

A g is smoother (as in small higher derivatives)
B All data points are globally strongly correlated
C Only close neighbors are correlated
D None of the above

Solution:
C. See GP slides 9.

Question 16 *(1 point)* You decide you want to use either $l = 0.1$ or $l = 0.5$. By choosing $l = 0.5$, what prior belief is expressed with this choice (as compared to 0.1)?

A g is smoother (as in small higher derivatives)

B All data points are globally strongly correlated

C Only close neighbors are correlated

D None of the above

Solution:
A. See GP slides 9.

Question 17 *(1 point)* Assume you choose $\sigma_F = 2$. What prior belief is expressed with this choice?

A g is rather smooth (as in small higher derivatives)

B All data points are globally strongly correlated

C Only close neighbors are correlated

D None of the above

Solution:
D. See GP slides 9. The lengthscale controls the distance over which data is assumed to be correlated. The output scale controls the magnitude of the function.

2.4 Cubic complexity reduction

Question 18 *(3 points)* You use a GP regression model to model a time series, i.e. inputs $t_i \in \mathbb{R}$ and outputs $x(t_i) \in \mathbb{R}$, with an RBF kernel $k(t_i, t_j) = \sigma_F^2 \exp\left(-\frac{1}{2l^2}(t_i - t_j)^2\right)$, with trainable hyperparameters σ_F, l. As the number of data points increase, you run into computational trouble, since both hyperparameter training and prediction time scale cubic in the number of observations. You decide that the kernel is the problem and want to substitute it. Which of the following changes will **NOT** help in reducing this cubic complexity, even when adapting the inference procedures accordingly?

A changing the RBF kernel to an approximation based on random Fourier features

B using inducing points instead of the full data set

C using a linear kernel, $k(t_i, t_j) = t_i t_j$, instead of the RBF kernel

D All of the above measures reduce the cubic complexity, if the inference schemes are adapted accordingly.

Solution:
D. All these methods were discussed in the lecture in the context of reducing computational complexity.

3 Variational Inference and MCMC

3.1 True/False questions

Question 19 *(1 point)* The KL-divergence $\mathrm{KL}(q||p)$ is well defined for the uniform distributions $p(x)$ and $q(x)$ shown in Figure 2.

A True B False

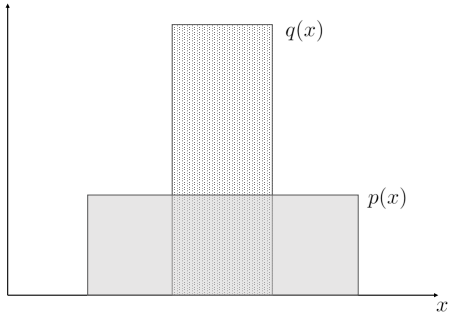


Figure 2: Distributions with different support

Solution:
True. By definition the KL-divergence is well-defined when the support of q is a subset of p's support.

Question 20 *(1 point)* Consider the probability density function $p(x)$ given in Figure 3. The Laplace approximation for this distribution will also be skewed.

A True B False

Solution:
False. Lecture 5, slide 17. The Laplace approximation does not work well when the true distribution is multi-modal or skewed, since the approximate distribution is a Gaussian, and therefore symmetric.

Question 21 *(1 point)* Consider $(X_N)_{N\geq 1}$ a random walk on an ergodic Markov chain that has a finite number of states. For every state x we have that $\lim_{N\to\infty} P(X_N = x)$ is not zero.

A True B False

Solution:
True. An ergodic Markov chain is reversible, i.e. every node can be reached from every other node, hence the probability of being in a node can not be zero.

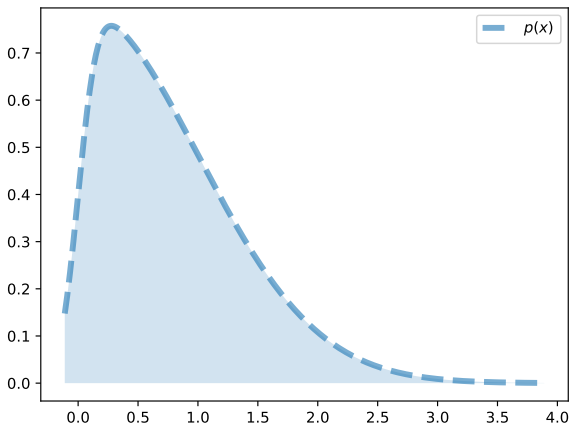
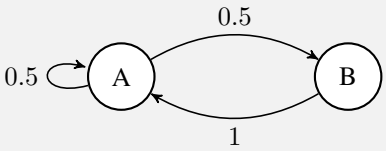


Figure 3: Related to Question 20

Question 22 *(1 point)* An ergodic Markov chain can have a state X from which transition to another state Z happens deterministically. In other words, the transition probability matrix of an ergodic Markov chain can have a row with only zeros and a one.

A True B False

Solution:
True. Lecture 6, slide 24. As an example. the following graph demonstrates an ergodic Markov chain with a deterministic transition.



3.2 Variational Inference

Question 23 *(3 points)* We want to use the Evidence Lower Bound to infer the parameters α, μ_1, μ_2 of a mixture of two Gaussian distributions (Figure 4). We denote the prior distribution on the parameters by $p(\alpha, \mu_1, \mu_2)$, the likelihood by

$$p(y|\alpha, \mu_1, \mu_2) = \alpha\mathcal{N}(y|\mu_1, 1) + (1-\alpha)\mathcal{N}(y|\mu_2, 1)$$

and the variational family by $\mathcal{Q}$. Assume there are n available samples $\{y_1, \cdots, y_n\}$ from this distribution. Calculate $\mathrm{ELBO}(q)$ the Evidence Lower Bound for some $q(\alpha, \mu_1, \mu_2) \in \mathcal{Q}$.

A $\sum_{i=1}^{n} \mathbb{E}_{\alpha,\mu_1,\mu_2\sim q} \log\left(\alpha e^{-(y_i-\mu_1)^2/2} + (1-\alpha)e^{-(y_i-\mu_2)^2/2}\right) - \frac{n}{2}\log 2\pi - \mathrm{KL}(q||p)$

B $\sum_{i=1}^{n} \mathbb{E}_{\alpha,\mu_1,\mu_2\sim q} \log\left(\alpha e^{-(y_i-\mu_1)^2/2} + (1-\alpha)e^{-(y_i-\mu_2)^2/2}\right) - \frac{n}{2}\log 2\pi + \mathrm{KL}(p||q)$

C $\mathbb{E}_{\alpha,\mu_1,\mu_2\sim q}\left[\sum_{i=1}^{n} \log\left(\frac{\alpha}{\sqrt{2\pi}}e^{-(y_i-\mu_1)^2/2} + \frac{(1-\alpha)}{\sqrt{2\pi}}e^{-(y_i-\mu_2)^2/2}\right) + \log p(\alpha,\mu_1,\mu_2) + \log q(\alpha,\mu_1,\mu_2)\right]$

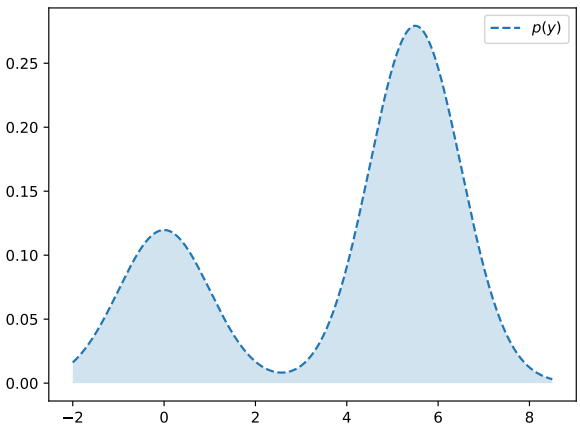


Figure 4: Related to Question 23

D $\mathbb{E}_{\alpha,\mu_1,\mu_2\sim q}\left[\sum_{i=1}^{n}\log\left(\frac{\alpha}{\sqrt{2\pi}}e^{-(y_i-\mu_1)^2/2}+\frac{(1-\alpha)}{\sqrt{2\pi}}e^{-(y_i-\mu_2)^2/2}\right)-\log q(\alpha,\mu_1,\mu_2)\right]$

Solution:
A. We have,

$$\begin{aligned}\mathrm{ELBO}(q) &= \mathbb{E}_{\boldsymbol{\theta}\sim q}\left[\log p(\boldsymbol{\theta}, y_{1:n}) - \log q(\boldsymbol{\theta})\right] \\ &= \mathbb{E}_{\boldsymbol{\theta}\sim q}\left[\log p(\boldsymbol{\theta}) + \sum_{i=1}^{n}\log p(y_i|\boldsymbol{\theta}) - \log q(\boldsymbol{\theta})\right] \\ &= -\mathrm{KL}(q||p) - \frac{n}{2}\log 2\pi + \sum_{i=1}^{n}\mathbb{E}_{\alpha,\mu_1,\mu_2\sim q}\log\left(\alpha e^{-(y_i-\mu_1)^2/2} + (1-\alpha)e^{-(y_i-\mu_2)^2/2}\right)\end{aligned}$$

Question 24 *(3 points)* Consider a poisson distribution, with unknown parameter λ,

$$p(k|\lambda) = \frac{\lambda^k e^{-\lambda}}{k!}.$$

We have observed 10 independent samples from this distribution,

$$D = \{0, 2, 2, 3, 2, 2, 2, 5, 3, 2\}.$$

Let $\hat{p}(k)$ denote the empirical distribution of D. Which λ minimizes the KL-distance between the poisson distribution and the empirical distribution $\mathrm{KL}\left(\hat{p}||p\right)$?

A 2
B 2.3
C 2.4
D The problem does not have a solution, since for every $\lambda \in (0, \infty)$ the KL-divergence is unbounded.

Solution:
B. Let $\hat{p}_k$ denote the empirical probability of k. Then the KL divergence between the true and the empirical distribution is

$$\begin{aligned}\mathrm{KL}(\hat{P}||P(\lambda)) &= \sum_k \hat{p}_k \left[\log(\hat{p}_k) - \log\left(p(k|\lambda)\right)\right] \\ &= \sum_k \hat{p}_k \left[\log(\hat{p}_k) - (k\log\lambda - \lambda - \log k!)\right]\end{aligned}$$

For $\hat{\lambda}$ that minimizes this distance we have,

$$\frac{\partial \mathrm{KL}(\hat{P}||P(\hat{\lambda}))}{\partial \lambda} = \sum_k \hat{p}_k - \frac{k\hat{p}_k}{\hat{\lambda}} = 0$$

which implies,

$$\hat{\lambda} = \frac{\sum_k k\hat{p}_k}{\sum_k \hat{p}_k} = \sum_k k\hat{p}_k.$$

Now from the dataset, we may calculate the $\hat{p}_k$,

k	0	2	3	5
$\hat{p}_k$	0.1	0.6	0.2	0.1

which, in turn, gives, $\hat{\lambda} = 2.3$.

3.3 Markov Chains and Gibbs Sampling

Consider a toy domain

$$D = \left\{ \begin{pmatrix} -1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\},$$

on which we define a stationary Markov chain with vector-valued states $\boldsymbol{Z} = (Z_1, Z_2) \in D$ and the stationary distribution $\boldsymbol{\pi}$.

Question 25 *(2 points)* Assume we have access to Q, the unnormalized joint stationary distribution, i.e., for every $(z_1, z_2) \in \{-1, 1\}^2$ we know the value $Q(Z_1 = z_1, Z_2 = z_2)$. At least how many operations are required to calculate $\mathbb{P}(Z_1 = 1|Z_2 = -1)$? Count the sums, multiplications, and divisions.

A 1 B 2 C 3 D 4

Solution:
B. Lecture 6, slide 33. This question hints at the fact that Gibbs updates require very little computations. We have,

$$\mathbb{P}(Z_1 = 1|Z_2 = -1) = \frac{Q(Z_1 = 1, Z_2 = -1)}{Q(Z_1 = 1, Z_2 = -1) + Q(Z_1 = -1, Z_2 = -1)}$$

therefore only 2 operations, one sum and one division, are required.

4 Bayesian Deep Learning and Approximate inference

In this questions we are modeling a *binary classification* problem with a 2-layer Bayesian neural network, where

$$P(y_i = 1) = \text{Bernoulli}(\sigma(\theta^\top \sigma(\mathbf{\Lambda} x_i))), \tag{1}$$

the function σ corresponds to sigmoid link function defined as

$$\sigma(x) = \frac{1}{1 + \exp(-x)}$$

acting on each component of a vector, and x_i are the features of the i-th datapoint. The network depends on two *unknown* parameters: θ and $\mathbf{\Lambda}$. Their respective dimension and prior information are summarized below.

$$\begin{aligned} \theta \in \mathbb{R}^2 \qquad & \theta \sim \mathcal{N}(0, \mathbf{I}_{2\times 2}) \\ \mathbf{\Lambda} \in \mathbb{R}^{2\times 3} \qquad & \mathbf{\Lambda}_{ij} \sim \mathcal{N}(0, 1), \end{aligned}$$

where all entries $\mathbf{\Lambda}_{ij}$ and θ are independent of each other.

4.1 Bayesian Deep Learning (8 points)

Question 26 *(1 point)* Aleatoric uncertainty of random variable y conditioned on x, corresponds to $\mathbb{E}_\theta\left(\text{var}\left(y|x, \theta, \mathbf{\Lambda}\right)\right)$. Here the expectation is over the prior distribution of θ.

A True B False

Solution:
False. The true aleatoric uncertainty averages over $\mathbf{\Lambda}$ as well, i.e. $\mathbb{E}_{\theta,\mathbf{\Lambda}}(\text{Var}(y|x, \theta, \mathbf{\Lambda}))$.

Question 27 *(4 points)* Given the dataset $\{x_i, y_i\}_{i=1}^{50}$, which of the optimization problems below describes the optimization problem to find maximum a posteriori estimate for the model parameters in Eq. (1)? Note that $\|\cdot\|_F$ denotes the Frobenius norm, while $\|\cdot\|_2$ denotes the spectral norm (the largest singular value of a matrix and for vectors l_2 norm).

A $\arg\min_{\theta,\mathbf{\Lambda}} \|\theta\|_2 + \|\mathbf{\Lambda}\|_F^2 - 2\sum_{i=1}^{50} y_i \log(\sigma(\theta^\top \sigma(\mathbf{\Lambda} x_i))) + (1 - y_i)\log(1 - \sigma(\theta^\top \sigma(\mathbf{\Lambda} x_i)))$

B $\arg\min_{\theta,\mathbf{\Lambda}} \|\theta\|_2 + \|\mathbf{\Lambda}\|_F^2 - \frac{1}{25}\sum_{i=1}^{50} y_i \log(\sigma(\theta^\top \sigma(\mathbf{\Lambda} x_i))) + (1 - y_i)\log(1 - \sigma(\theta^\top \sigma(\mathbf{\Lambda} x_i)))$

C $\arg\min_{\theta,\mathbf{\Lambda}} \|\theta\|_2 + \|\mathbf{\Lambda}\|_F^2 - \frac{1}{25}\sum_{i=1}^{50} y_i \log(\sigma(\theta^\top \sigma(\mathbf{\Lambda} x_i)))$

D $\arg\min_{\theta,\mathbf{\Lambda}} \|\theta\|_2 + \|\mathbf{\Lambda}\|_F^2 - 2\sum_{i=1}^{50} y_i \log(\sigma(\theta^\top \sigma(\mathbf{\Lambda} x_i)))$

E $\arg\min_{\theta,\mathbf{\Lambda}} \|\theta\|_2 + \|\mathbf{\Lambda}\|_2^2 - 2\sum_{i=1}^{50} y_i \log(\sigma(\theta^\top \sigma(\mathbf{\Lambda} x_i))) + (1 - y_i)\log(1 - \sigma(\theta^\top \sigma(\mathbf{\Lambda} x_i)))$

Solution:
A. The solution follows by calculating the likelihood times the prior probability and taking the negative log. The prior is with l_2-norm for both matrix and the vector, which in the case of matrices is the Frobenius norm.

Question 28 *(1 points)* We choose to approximate the posterior by Langevin sampling. Which of the below represents a correct update for θ of Langevin sampling? In the options below, $L(\theta, \mathbf{\Lambda})$ denotes the likelihood $p((x_i, y_i)_{i=1}^{50}|\theta, \mathbf{\Lambda})$. At iteration k we sample $\varepsilon_k \sim \mathcal{N}(0, 2\eta_k)$.

A $\theta_{k+1} = \theta_k - \eta_k(-\theta_k + \nabla_\theta \log(L(\theta_k, \mathbf{\Lambda}_k))) + \varepsilon_k$
B $\theta_{k+1} = \theta_k - \eta_k(\nabla_\theta \log(L(\theta_k, \mathbf{\Lambda}_k))) + \varepsilon_k$
C $\theta_{k+1} = \theta_k + \eta_k(-\theta_k + \nabla_\theta \log(L(\theta_k, \mathbf{\Lambda}_k))) + \varepsilon_k$
D $\theta_{k+1} = \theta_k + \eta_k(\nabla_\theta \log(L(\theta_k, \mathbf{\Lambda}_k))) + \varepsilon_k$

Solution:
C. It follows from the definition of Langevin dynamics.

4.2 Prediction and Calibration (4 points)

Question 29 *(3 points)* By performing approximate inference, we can sample parameters from the approximate posterior. We obtained three samples below for a probing value $x = \bar{x}$:

$$\sigma(\mathbf{\Lambda}_1\bar{x}) = (0,1) \quad \theta_1 = (2,1) \qquad \sigma(\mathbf{\Lambda}_2\bar{x}) = (0.5,0.5) \quad \theta_2 = (1,1) \qquad \sigma(\mathbf{\Lambda}_3\bar{x}) = (0,1) \quad \theta_3 = (1,0)$$

Calculate the approximate expected value of $y(\bar{x})$ over the predictive distribution $P(y(\bar{x})|\{x_i, y_i\}_{i=1}^{50})$ based on the above samples.

A 0.23 B 0.5 C 0.65 D 0.04

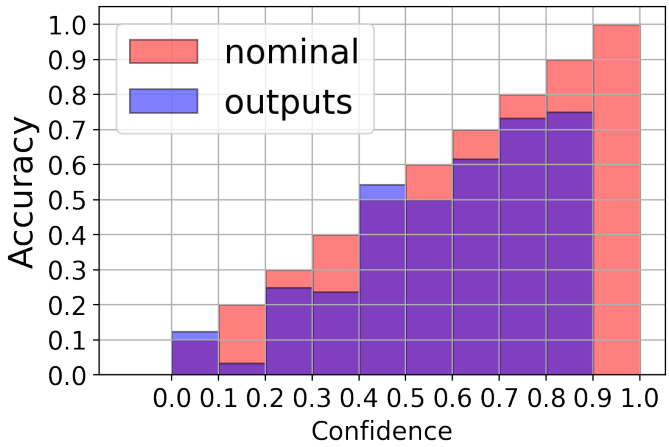
Solution:
C. In order to calculate expected predictive distribution we need to calculate the predicted value for each and then average.

$$\mathbb{E}[y(\bar{x})] \approx \frac{1}{3}\left(\sigma((0,1)^\top(2,1)) + \sigma((0.5,0.5)^\top(1,1)) + \sigma((0,1)^\top(1,0))\right) \quad (2)$$
$$= \frac{1}{3}\left(\sigma(1) + \sigma(1) + \sigma(0)\right) \approx 0.65 \quad (3)$$

The other solutions correspond to wrong way to average.

Question 30 *(3 points)* We would like to understand the calibration of our approximate predictive distribution on a small holdout set.

In the lectures you explored the use *reliability diagrams* as means to do it. Which of the statements bellow summarizes the information in the depicted reliability diagram the best?



A The model is well-calibrated except for when model is very confident between $[0.9, 1]$.

B The model is reasonably calibrated across all confidence levels. Worst calibration occurs in $[0.1, 0.2]$ and $[0.8, 0.9]$, where the model is overconfident.

C The model is over-confident in its prediction for all confidence levels.

D The model is reasonably calibrated across all confidence levels. Worst calibration occurs in $[0.1, 0.2]$ and $[0.8, 0.9]$, where the model is better than it predicts.

Solution:
B.

1. A is wrong since, it is possible that there are missing outputs that which have confidence above 0.9.
2. C is not true since between $[0, 0.1]$ and $[0.4, 0.5]$ we can't draw this conclusion.
3. D is wrong since the accuracy in $[0.1, 0.2]$ is lower than that one would expect
4. This leaves B as the most plausible explanation.

5 Reinforcement Learning

5.1 Markov Decision Processes

Consider a Markov decision process (MDP) with six states $\mathcal{X} = \{A, B, C, D, E, F\}$ and four actions $\mathcal{A} = \{\leftarrow, \uparrow, \rightarrow, \downarrow\}$, representing a two dimensional grid, shown below:

A	B	C
D	E	F

The agent always starts in state $x_0 = A$, and it can take an action to move in four directions. If the agent tries to move in a direction with no adjacent state, it will stay in its previous state. If there is an adjacent state, then with probability $1 - \delta$, the agent moves to that state, and with probability $\delta \in [0, 1]$, it stays in its previous state. We will consider different reward functions for this environment, but the rewards only depend on the state the agent ends up in after a transition, i.e., $r(x, a, x') = r(x')$.

The agent's goal is to find an optimal policy, i.e., a policy that maximizes

$$J(\pi) = \mathbb{E}\left[\sum_{t=0}^{T-1} \gamma^t r(x_{t+1})\right]$$

where T is the episode length and the expectation is computed over trajectories $\tau = (x_0, \ldots, x_T)$ generated by the policy π and the transition dynamics of the environment. With γ we denote the discount factor.

Question 31 *(1 points)* For $\delta = 1$, we can find a deterministic policy $\pi^* : \mathcal{X} \to \mathcal{A}$ that is optimal for all possible reward functions $r : \mathcal{X} \to \mathbb{R}$ and discount factors $\gamma \in (0, 1)$.

A True B False

Solution:
True. If $\delta = 1$, the agent always stays in state A and it does not matter which action it takes. Therefore every policy is optimal independent of the reward function and the discount factor.

Question 32 *(1 points)* For $\delta = 0$, the environment has only deterministic dynamics.

A True B False

Solution:
True. For $\delta = 0$ the agent always moves in the direction of the action it takes except if it is at the boundary of the grid. There is no randomness, hence the dynamics are deterministic.

Question 33 *(3 points)* Assume the reward function is given by $r(A) = r(B) = r(C) = r(E) = 0$ and $r(D) = 0.15$ and $r(F) = 1$. Assume a fixed episode length of $T = 3$ steps, $\delta = 0$, and discount factor $\gamma = 0.5$. After the last timestep an agent that follows an optimal policy will be in:

A State B
B State C
C State D
D State E
E State F

Solution:
F. First, note that the agent can stay in any state by choosing an action that would move outside the grid. Only the rewards of state D and F are non-zero and the dynamics are deterministic ($\delta = 0$). Therefore, it will either be optimal for the agent to move the state D and stay there (trajectory $\tau_1 = (A, \downarrow, D, \downarrow, D, \downarrow, D)$) or move through state D to state F (trajectory $\tau_2 = (A, \downarrow, D, \rightarrow, E, \rightarrow, F)$). We can compute the return of both sequences:

$$J(\tau_1) = r(D) + \gamma \cdot r(D) + \gamma^2 \cdot r(D) = 0.15 + 0.5 \cdot 0.15 + 0.25 \cdot 0.15 = 0.2625$$
$$J(\tau_2) = r(D) + \gamma \cdot r(E) + \gamma^2 \cdot r(F) = 0.15 + 0 + 0.25 \cdot 1 = 0.4$$

As $r(\tau_2) > r(\tau_1)$ any optimal policy will follow trajectory τ_2 and therefore the agent will end up in state F.

Question 34 *(2 points)* The transition dynamics of this MDP can be described using four transition matrices $P_\leftarrow, P_\uparrow, P_\rightarrow$, and $P_\downarrow$ for taking action $\leftarrow, \uparrow, \rightarrow$, and $\downarrow$ respectively. The matrix rows denote the source state, the columns the target state and states are ordered alphabetically. What is the transition matrix $P_\uparrow$, i.e., if the agent always chooses the action $\uparrow$?

A $P_\uparrow = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ \delta & 0 & 0 & 1-\delta & 0 & 0 \\ 0 & \delta & 0 & 0 & 1-\delta & 0 \\ 0 & 0 & \delta & 0 & 0 & 1-\delta \end{pmatrix}$

B $P_\uparrow = \begin{pmatrix} \delta & 0 & 0 & 1-\delta & 0 & 0 \\ 0 & \delta & 0 & 0 & 1-\delta & 0 \\ 0 & 0 & \delta & 0 & 0 & 1-\delta \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$

C $P_\uparrow = \begin{pmatrix} 1-\delta & 0 & 0 & \delta & 0 & 0 \\ 0 & 1-\delta & 0 & 0 & \delta & 0 \\ 0 & 0 & 1-\delta & 0 & 0 & \delta \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$

D $P_\uparrow = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1-\delta & 0 & 0 & \delta & 0 & 0 \\ 0 & 1-\delta & 0 & 0 & \delta & 0 \\ 0 & 0 & 1-\delta & 0 & 0 & \delta \end{pmatrix}$

Solution:
D. In states A, B, C moving up will cause the agent to stay in the same state; hence, the transition matrix $P_\uparrow$ needs to have a 3×3 unit matrix in the top left and zeros in the top right. The last three rows describe the agent starting in states D, E, F. In that case the agent will stay in the same state with probability δ and move to the states A, B, C with probability $1 - \delta$ respectively. Therefore, $P_\uparrow$ needs to have δ on the rest of the main diagonal and a diagonal matrix with entries $1 - \delta$ in the bottom left. The correct answer is D.

For the following two exercises, assume that the reward function is given by $r(A) = r(B) = r(C) = 1$ and $r(D) = r(E) = r(F) = 2$.

Question 35 *(4 points)* Now, assume the episode length is $T = 2$ with a discount factor $\gamma = 1$, and $\delta = \frac{1}{4}$. Consider a deterministic policy $\pi : \mathcal{X} \to \mathcal{A}$ with $\pi(A) = \rightarrow, \pi(B) = \downarrow, \pi(C) = \rightarrow, \pi(D) = \uparrow, \pi(E) = \leftarrow, \pi(F) = \leftarrow$. Compute $J(\pi)$.

A $J(\pi) = \frac{33}{16}$
B $J(\pi) = \frac{41}{16}$
C $J(\pi) = \frac{31}{64}$
D $J(\pi) = \frac{41}{64}$
E $J(\pi) = \frac{5}{12}$

Solution:
B. We can list all possible trajectories $\tau = (x_0, a_0, x_1, a_1, x_2)$ with probabilities $P(\tau) = P(x_1|x_0, a_0)P(x_2|x_1, a_1)$ and return $J(\tau) = r(x_1) + r(x_2)$. The policy is deterministic and does not affect the probability of a trajectory. There are 4 possible trajectories:

$$\tau_1 = (A, \rightarrow, A, \rightarrow, A); P(\tau_1) = \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}; J(\tau_1) = 2$$
$$\tau_2 = (A, \rightarrow, A, \rightarrow, B); P(\tau_2) = \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{16}; J(\tau_2) = 2$$
$$\tau_3 = (A, \rightarrow, B, \downarrow, B); P(\tau_3) = \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{16}; J(\tau_3) = 2$$
$$\tau_4 = (A, \rightarrow, B, \downarrow, E); P(\tau_4) = \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{16}; J(\tau_4) = 3$$

Overall, we can compute

$$J(\pi) = \sum_{i=1}^{4} P(\tau_i)J(\tau_i) = \frac{1}{16} \cdot 2 + \frac{3}{16} \cdot 2 + \frac{3}{16} \cdot 2 + \frac{9}{16} \cdot 3 = \frac{41}{16}$$

Question 36 *(2 points)* So far, we considered a fixed episode length T and deterministic policies. Instead, we can also consider the *infinite-horizon* setting, where

$$J(\pi) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r(x_{t+1})\right]$$

and stochastic policies π, where $\pi(a|s)$ denotes the probability of taking action a in state x. Consider a stochastic policy π with $\pi(\uparrow |x) = \frac{3}{4}$, $\pi(\downarrow |x) = \frac{1}{4}$ and $\pi(\leftarrow |x) = \pi(\rightarrow |x) = 0$ for all $x \in \mathcal{X}$. This policy randomizes between going up and down and never goes left or right. Assume the discount factor is $\gamma = 0.5$ for an infinite time horizon, and $\delta = 0$. Compute $J(\pi)$.

A $J(\pi) = 1$
B $J(\pi) = 1.5$
C $J(\pi) = 2$
D $J(\pi) = 2.5$

Solution:
D. Denote the value function of a policy π as state x by $V^\pi(x)$. Then $J(\pi) = V^\pi(A)$. Consider the Bellman equation which in our case becomes

$$V^\pi(A) = \pi(\uparrow |A)(r(A) + \gamma V^\pi(A)) + \pi(\downarrow |A)(r(D) + \gamma V^\pi(D))$$
$$V^\pi(D) = \pi(\uparrow |A)(r(A) + \gamma V^\pi(A)) + \pi(\downarrow |A)(r(D) + \gamma V^\pi(D))$$

We can observe that in this case actually $V^\pi(A) = V^\pi(D)$ and

$$V^\pi(A) = \frac{3}{4} + \frac{3}{8}V^\pi(A) + \frac{1}{2} + \frac{1}{8}V^\pi(A) = \frac{1}{2}V^\pi(A) + \frac{5}{4}$$

It follows that $J(\pi) = V^\pi(A) = 2.5$.

For the rest of the exercises, let $\delta = 0.1$ and the reward function be $r(A) = 0.5, r(B) = 0, r(C) = 1, r(D) = 0.5, r(E) = 0, r(F) = 1$, and discount factor $\gamma = 0.9$.

We now want to run optimistic Q-learning with a fixed learning rate $\alpha_t = 0.5$ to find a good policy in the environment. Recall, that optimistic Q-learning initializes Q-values with

$$Q(x,a) = \frac{R_{\text{max}}}{1-\gamma}\prod_{t=1}^{T_{\text{init}}}(1-\alpha_t)^{-1}$$

and at time t takes action $a_t = \arg\max_a Q(x_t, a)$.

Question 37 *(1 points)* Is this approach guaranteed to converge to the optimal Q value?

A True B False

Solution:
False. To guarantee convergence we need to decay the learning rate.

Question 38 *(1 point)* Let $R_{\text{max}} = 2$, and $T_{\text{init}} = 2$. How will optimistic Q-learning initialize the Q-value estimates?

A 1
B 50
C 10
D 80
E 100

Solution:

$$Q(x,a) = \frac{R_{\text{max}}}{1-\gamma}\prod_{t=1}^{T_{\text{init}}}(1-\alpha)^{-1} = \frac{2}{0.1}(1-0.5)^{-2} = 20 \cdot 4 = 80$$

The first two transitions you observe are:

$$(x = A, a = \rightarrow, r = 0, x' = B)$$
$$(x = B, a = \rightarrow, r = 0, x' = B)$$

Question 39 *(2 points)* What is $Q(B, \rightarrow)$ after updating for these two transitions?

A 4
B 4.25
C 4.5
D 4.75
E 5

Solution:
D. We use the formula for Q updates:

$$Q(x,a) \leftarrow (1-\alpha_t)Q(x,a) + \alpha_t(r + \gamma \max_a Q(x', a')).$$

Initially all $Q(x,a) = 5$. The first transition reduces $Q(A, \rightarrow)$ but does not affect $Q(B, \rightarrow)$, the second transition updates:

$$Q(B, \rightarrow) = 0.5 \cdot 5 + 0.5(0 + 0.9 \cdot 5) = 4.75$$

Question 40 *(2 points)* After running Q-learning for a while, the Q-value estimates for state A are:

$$\begin{aligned} Q(A, \rightarrow) &= 1 \\ Q(A, \uparrow) &= 2 \\ Q(A, \leftarrow) &= 3 \\ Q(A, \downarrow) &= 1.5 \end{aligned}$$

After starting in state $x_t = A$, taking a greedy action a_t, and updating the Q-values, what is the values of $Q(A, \leftarrow)$ and $Q(A, \rightarrow)$?

A $Q(A, \leftarrow) = 3.2, Q(A, \rightarrow) = 1.5$
B $Q(A, \leftarrow) = 3.1, Q(A, \rightarrow) = 1.1$
C $Q(A, \leftarrow) = 3.2, Q(A, \rightarrow) = 1$
D $Q(A, \leftarrow) = 3.1, Q(A, \rightarrow) = 1$

Solution:
D. The greedy policy will take action $\leftarrow$ in state A, so it will not update $Q(A, \rightarrow)$. Taking action $\leftarrow$ leads to state A with reward $r = 0.5$, resulting in:

$$\begin{aligned} Q(A, \leftarrow) \leftarrow (1 - \alpha_t) Q(A, \leftarrow) + \alpha_t (r + \gamma \max_a Q(A, \leftarrow)) &= \frac{1}{2} \cdot 3 + \frac{1}{2} \left(\frac{1}{2} + 0.9 \cdot 3 \right) \\ &= 1.5 + 0.25 + 1.35 = 3.1 \end{aligned}$$

Question 41 *(1 points)* After running Q-learning for even longer, the Q-value estimates for state A have converged to:

$$\begin{aligned} Q(A, \rightarrow) &= 1 \\ Q(A, \uparrow) &= 2.5 \\ Q(A, \leftarrow) &= 2.5 \\ Q(A, \downarrow) &= 1.5 \end{aligned}$$

Assuming that the algorithm converged to the optimal Q-function, what is the value function $V^{\pi^*}(A)$ of the optimal policy π^*?

A $V^{\pi^*}(A) = 1.25$
B $V^{\pi^*}(A) = 1.5$
C $V^{\pi^*}(A) = 2$
D $V^{\pi^*}(A) = 2.5$

Solution:

$$V^{\pi^*}(s) = \mathbb{E}_{a \sim \pi^*} Q^*(s, a) = \max Q^*(s, a)$$

Assuming that Q-learning converged to Q^*, we have $V^{\pi^*}(A) = 2.5$.

5.2 True/False question

Question 42 *(1 point)* Value iteration finds the optimal policy in a Markov decision process (MDP) with a number of iterations polynomial in the size of the state and action space.

A True B False

Solution:
False. Value iteration is guaranteed to converge to an ε-optimal policy, but it is not guaranteed to find the exactly optimal policy. The statement is therefore false.

Question 43 *(1 point)* Policy iteration monotonically improves the associated value function in each step.

A True B False

Solution:
True. Policy iteration is guaranteed to monotonically improve the policy.

Question 44 *(1 point)* We can calculate the optimal policy in a partially observable Markov decision process (POMDP), where the horizon, action space, and state space are all finite, using dynamic programming.

A True B False

Solution:
True. By representing a POMDP as a belief-state MDP, we can use any method for solving MDPs, including dynamic programming.

Question 45 *(1 point)* Deep Deterministic Policy Gradient (DDPG) injects random noise into the policy to ensure sufficient exploration.

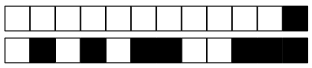
A True B False

Solution:
True. DDPG randomly perturbs the policy during exploration.

Question 46 *(1 point)* In practice, current model-based deep RL techniques can often help to significantly reduce the sample complexity of finding a well-performing policy compared to model-free techniques.

A True B False

Solution:
True. If we have an accurate model of the environment, we can use it for planning, i.e., to find a good policy. It is often significantly more sample efficient to learn a model of the environment to do planning in, compared to directly learning a policy in a model-free way.



+1/21/39+

Answer Sheet of the Probabilistic Artificial Intelligence Exam

0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9

⟵ Please encode your student number on the left, and write your first and last names below.

Firstname and Lastname:

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Question 1: A B C D
Question 2: A B C D
Question 3: A B
Question 4: A B
Question 5: A B
Question 6: A B
Question 7: A B
Question 8: A B C D
Question 9: A B C D
Question 10: A B C D
Question 11: A B C D
Question 12: A B C D
Question 13: A B C D
Question 14: A B C D
Question 15: A B C D
Question 16: A B C D
Question 17: A B C D
Question 18: A B C D
Question 19: A B
Question 20: A B
Question 21: A B
Question 22: A B
Question 23: A B C D
Question 24: A B C D
Question 25: A B C D
Question 26: A B
Question 27: A B C D E
Question 28: A B C D
Question 29: A B C D
Question 30: A B C D
Question 31: A B
Question 32: A B
Question 33: A B C D E
Question 34: A B C D
Question 35: A B C D E
Question 36: A B C D
Question 37: A B
Question 38: A B C D E
Question 39: A B C D E
Question 40: A B C D
Question 41: A B C D
Question 42: A B
Question 43: A B
Question 44: A B
Question 45: A B
Question 46: A B

For your examination, preferably print documents compiled from auto-multiple-choice.